faculty of science
department of mathematics

Hadia Niaz

MATH 100, Spring 2026
Final Exam
April 20, 2026, 12:00 – 15:00

SFU Email:	hna58@SFU.CA	Signature:	hn

First name (please write as legibly as possible within the boxes)

HADIA

Last name

NIAZ

Student number (9 digits)

301633056

Please Read the Following Instructions

1. Do not open this booklet until told to do so.
2. **Using blockletters and dark pen,** write your name, SFU student number, and email ID in the space provided.
3. This exam is **180 minutes** long and consists of **20 questions**.
4. You may use the last page for extra space. **You must indicate below the question if any of your work is written elsewhere.**
5. To receive full credit for written questions, your solution must be complete and well presented. You must cross out any work that you do not wish to be graded.
6. **Calculators are not allowed.** You should leave answers as "calculator ready" expressions (such as $\sqrt{3}$ or π); however, answers should be simplified as much as possible.
7. No books, notes, smartphones, or any device capable of connecting to the internet shall be within the reach of a student during the examination.
8. **During the examination, copying from, communicating with, or deliberately exposing written papers to the view of other examinees is forbidden.**

Question:	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Total
Points:	3	4	3	5	4	4	5	4	3	4	5	4	3	4	3	5	4	4	4	5	80

Potentially Useful Identities and Formulas

Addition and Subtraction Trigonometric Identities:

$$\cos(\alpha+\beta) = \cos(\alpha)\cos(\beta) - \sin(\alpha)\sin(\beta)$$
$$\cos(\alpha-\beta) = \cos(\alpha)\cos(\beta) + \sin(\alpha)\sin(\beta)$$
$$\sin(\alpha+\beta) = \sin(\alpha)\cos(\beta) + \cos(\alpha)\sin(\beta)$$
$$\sin(\alpha-\beta) = \sin(\alpha)\cos(\beta) - \cos(\alpha)\sin(\beta)$$

Double Angle Trigonometric Identities:

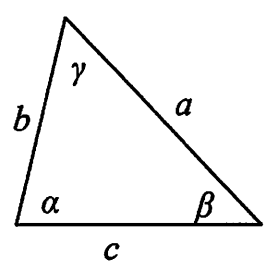
$$\sin(2\theta) = 2\sin(\theta)\cos(\theta)$$
$$\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta) = 1 - 2\sin^2(\theta) = 2\cos^2(\theta) - 1$$

Half Angle Trigonometric Identities:

$$\sin\left(\frac{\alpha}{2}\right) = \pm\sqrt{\frac{1-\cos(\alpha)}{2}}$$
$$\cos\left(\frac{\alpha}{2}\right) = \pm\sqrt{\frac{1+\cos(\alpha)}{2}}$$

Laws of Sines and Cosines:

Consider a triangle with angles and opposite sides as shown:



$$\frac{\sin(\alpha)}{a} = \frac{\sin(\beta)}{b} = \frac{\sin(\gamma)}{c}$$

$$a^2 = b^2 + c^2 - 2bc\cos(\alpha)$$
$$b^2 = a^2 + c^2 - 2ac\cos(\beta)$$
$$c^2 = a^2 + b^2 - 2ab\cos(\gamma)$$

[3] 1. A line passes through $(1, 4)$ and is perpendicular to $3x - 2y = 6$. Find the equation of this line.

$3(1) - 2(4) = 6$

$3 - 8 = 6$

$-5 = 6$

$3x - 2y = 6$

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[4] 2. Let $f(x) = \sqrt{x-3}$ and $g(x) = x^2 + 1$. Find the simplified expression for $(g \circ f)(x)$ and state its domain in interval notation.

$(g \circ f)(x) = \sqrt{(x^2+1) - 3}$

$(-\infty, -3)$

[3] 3. Starting with the parent function $f(x) = |x|$, write the equation for $g(x)$ that results from a horizontal shift right by 2 units, a vertical stretch by a factor of 3, and a reflection over the x-axis.

$f(x) = |x|$

$f(x) = |(x+2)|$

$f(x) = -|(x+2)|$

$f(x) = -3|(x+2)|$

right 2 units

vertical stretch by 3

reflection over x axis

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[5] 4. Find the inverse $f^{-1}(x)$ for $f(x) = \dfrac{2x-5}{3x+1}$.

$y = \dfrac{2x-5}{3x+1}$

$x = \dfrac{2y-5}{3y+1}$

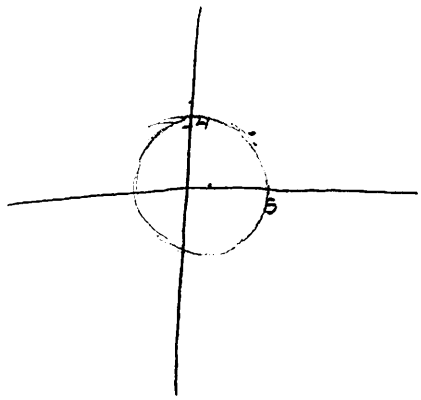
$x = \dfrac{2-5}{3+1}$

$x = \dfrac{-3}{3}$

$x = -3$

$f^{-1}(x) = -3$

[4] 5. Write the equation of the circle in standard form with centre $(1, -1)$ passing through the point $(5, 4)$.



[4] 6. Factor $P(x) = x^3 - 4x^2 - 7x + 10$ completely, given that $(x - 1)$ is one of its factors. State all real zeros.

$P(x) = (x^2 + 5)(x-2)(x-1)$

zeros: $1, 2, -5$

[5] 7. For the function $R(x) = \dfrac{2x^2 - 8}{x^2 - x - 6}$, identify any holes, vertical asymptotes, and horizontal asymptotes.

VA: -8

HA: -6

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[4] 8. A population of bacteria doubles every 4 hours. If the initial population is 500, write an exponential model for $P(t)$, where t is measured in hours. Leave your answer in calculator-ready format.

$P(t) = 500$

$\log_4 \boxed{}$

$4^{\boxed{t}}$

[3] 9. Use properties of logarithms to expand $\log\left(\frac{x^3\sqrt{y}}{z^2}\right)$. Express in terms of $\log(x)$, $\log(y)$, and $\log(z)$.

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[4] 10. Solve for x:

$$\ln(x) + \ln(x-3) = \ln(4)$$

$e^{\square} = e^{\square}$

$\ln(x)$

[5] 11. A car bought for \$25,000 depreciates at a rate of 12% per year. After how many years will the car be worth exactly \$10,000? Express your answer in calculator-ready format.

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[4] 12. Find the exact length of an arc intercepted by a central angle of 135° in a circle with a radius of 8 cm.

[3] 13. A point P is on a circle of radius 5 at an angle of $\theta = \frac{5\pi}{6}$. Determine the exact (x, y) coordinates of point P.

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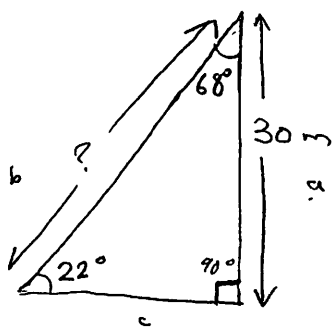
[4] 14. From the top of a 30 m lighthouse, the angle of depression to a boat in the ocean is 22°. How far is the boat from the base of the lighthouse? Leave your answer in calculator-ready format.

$$\frac{\sin(22)}{30} = \frac{\sin(90^\circ)}{b}$$



$$\begin{array}{r} 90^\circ \\ +\,22^\circ \\ \hline 112^\circ \end{array} \qquad \begin{array}{r} 1\overset{7}{8}\overset{10}{0} \\ -112 \\ \hline 068 \end{array}$$

[3] 15. Simplify the expression $(\sec\theta - \tan\theta)(\sec\theta + \tan\theta)$ using fundamental identities.

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[5] 16. State the amplitude, period, phase shift, and midline for $f(x) = -4\sin(2x - \pi) + 3$.

amplitude: 4
period: 8

midline: 3

[4] 17. Find the exact value of $\sin\left(\cos^{-1}\left(-\frac{1}{2}\right)\right)$.

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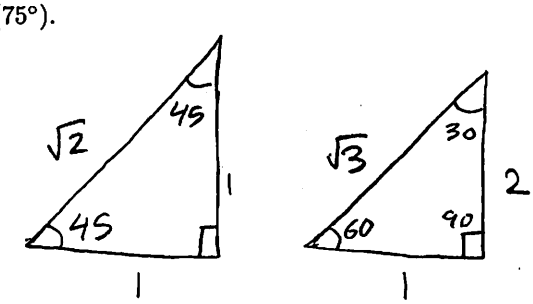
[4] 18. Use an addition or subtraction identity to find the exact value of $\cos(75°)$.

$\cos(75°)$

SOH CAH TOA



$$\cos(45+30) = \cos(45)\cos(30) - \sin(45) - \sin(30)$$

$$= \left(\frac{1}{\sqrt{2}}\right)\left(\frac{2}{\sqrt{3}}\right) - \left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{3}}\right)$$

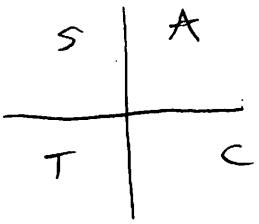
$$= \frac{2}{\sqrt{6}} - \frac{1}{\sqrt{6}}$$

$$\cos(75°) = \frac{1}{\sqrt{6}}$$

[4] 19. If $\sin\theta = \frac{3}{5}$ and θ is in Quadrant II, find the exact value of $\sin(2\theta)$.

$\sin 2\theta = 2\sin(\theta)\cos(\theta)$

$\sin 2\theta = 2\sin\left(\frac{3}{5}\right)\cos\left(\frac{3}{5}\right)$



[5] 20. In triangle ABC, side $a = 10$, side $b = 12$, and the contained angle $C = 60°$. Find the exact length of side c.

$c^2 = a^2 + b^2 - 2ab\cos(\gamma)$

$c^2 = 10^2 + 12^2 - 2(10)(12)\cos(60°)$

$c^2 = 100 + 144 - 2(120)\cos(60°)$

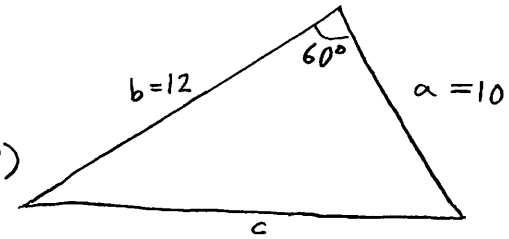
$c^2 = 244 - 240\cos(60°)$

$c^2 = 4\left(\frac{10}{12}\right)$

$c^2 = 4\left(\frac{5}{6}\right)$

$c^2 = \frac{20}{6}$

$c = \sqrt{\frac{10}{2}}$



$= \sqrt{\frac{10}{2}}$

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